Neck: triangles, fascia & nodes

Surface landmarks, the triangles and their contents, deep cervical fascia, the neck spaces, and the lymph node levels you need for neck dissection.

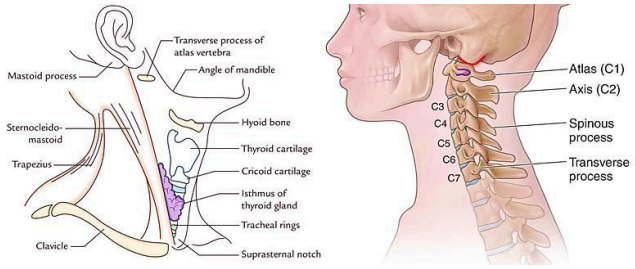
1. Surface landmarks
2. Sternocleidomastoid and trapezius
3. Triangles of the neck
4. Posterior triangle in detail
5. Deep cervical fascia
6. Neck spaces
7. Lymph node groups
8. Lymph node levels
9. In theatre
10. On imaging

01 Surface landmarks

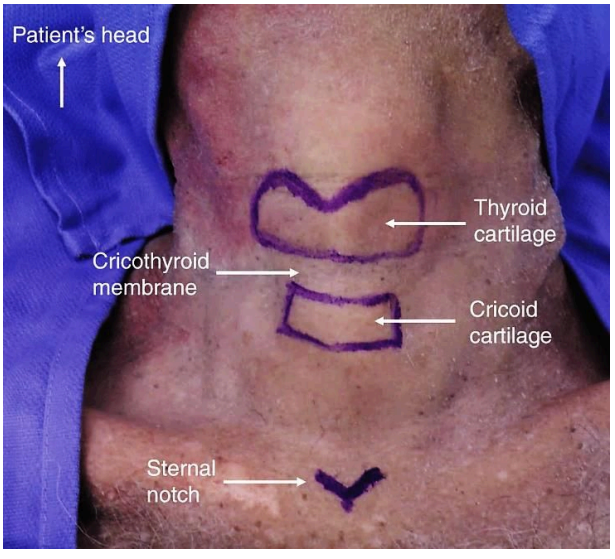
- **Hyoid** (C3): thyroglossal cyst moves up on tongue protrusion.
- **Thyroid cartilage** (Adam's apple, C4–C5) and **cricoid** (C6), with the **cricothyroid membrane** between them.
- **Thyroid isthmus** is palpable below the cricoid over rings 2–4; normal lobes lie deep to SCM.
- **Carotid bifurcation** at the upper border of the thyroid cartilage, where the carotid pulse is best felt.
- **Mastoid**: transverse process of the atlas is felt just in front of and below it.

CLINICAL PEARL C6 marks many things: cricoid, end of larynx and pharynx, start of trachea and oesophagus, Chassaignac's tubercle, vertebral artery entering the foramen transversarium.

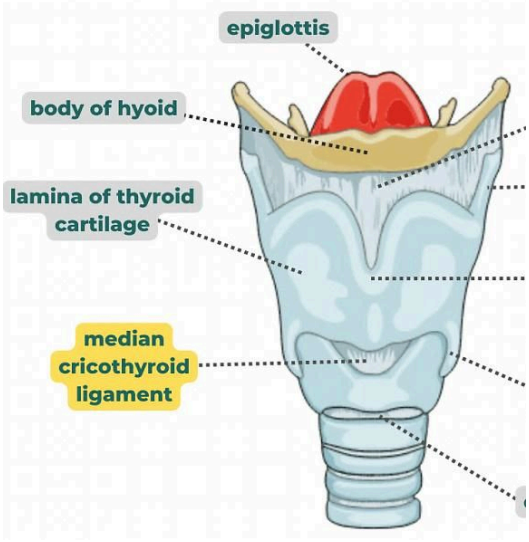
SOURCES DIFFER CAROTID BIFURCATION: ENT Steps: level of the greater cornu of the hyoid. Doss: upper border of the thyroid cartilage (C3/C4 disc). **HYOID:** Doss C2–C3; most texts C3.



Surface landmarks of the neck
ENT Steps Anatomy, p. 260

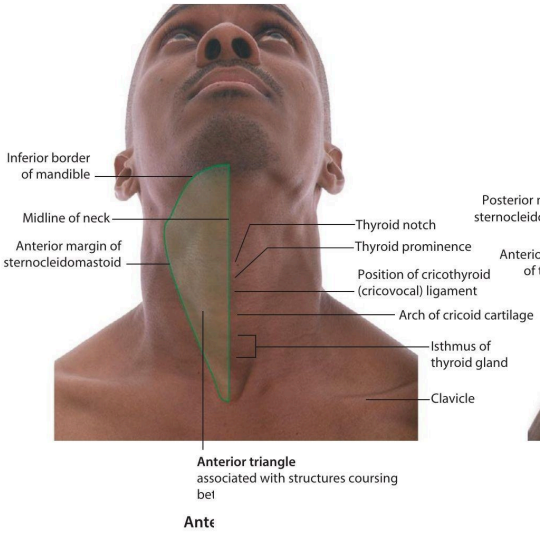


Surface marking of the thyroid and cricoid cartilages and cricothyroid membrane
ENT Steps Anatomy, p. 262



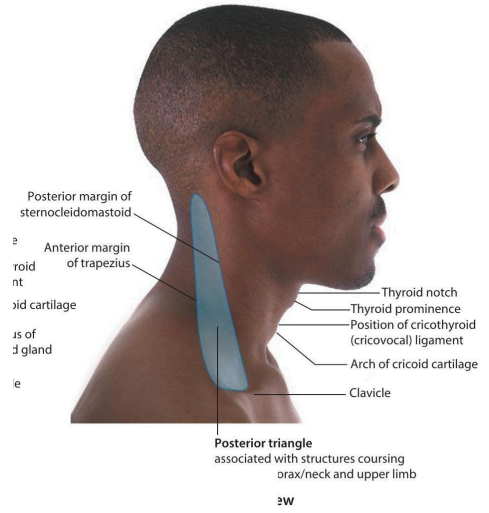
Laryngeal framework for surface anatomy

ENT Steps Anatomy, p. 262



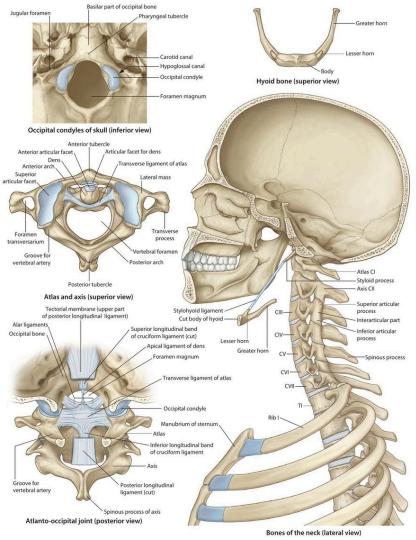
Neck surface anatomy (anterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 536



Neck surface anatomy (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 536



Bones of the neck: hyoid, atlas, axis and cervical spine

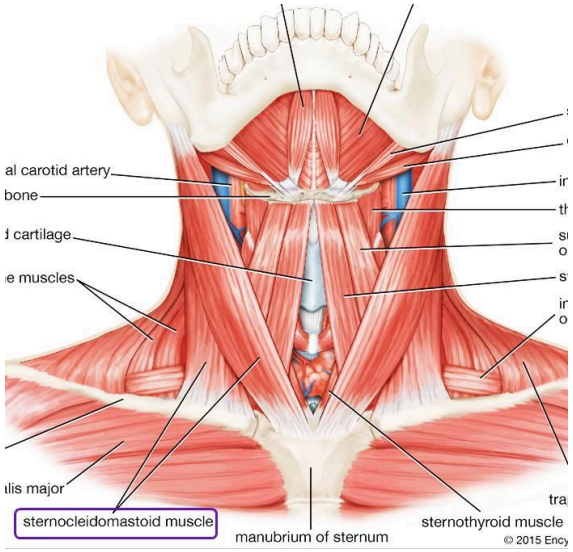
Gray's Atlas of Anatomy, 2nd ed., p. 537

02 Sternocleidomastoid and trapezius

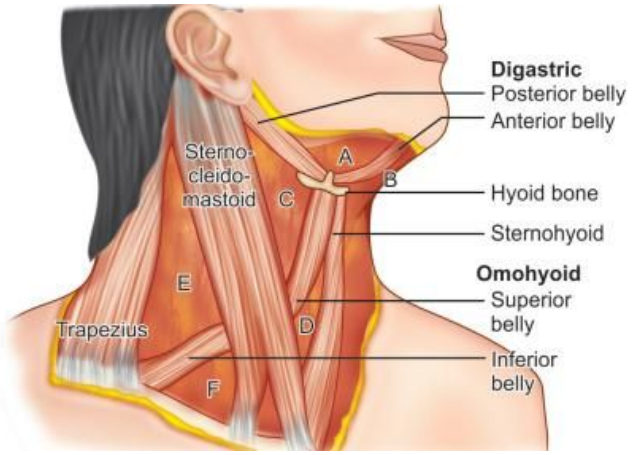
- **SCM:** sternal head (manubrium) and clavicular head (medial third) to the mastoid and superior nuchal line. **Accessory nerve** motor; C2–C3 proprioception (C2–C4 in ENT Steps).
- One side tilts the head to the same side and turns the face to the opposite side; both sides flex the neck.
- The **IJV** lies in the gap between the two heads; the carotid sheath and deep jugular nodes lie under the muscle.
- **Erb's point:** junction of upper and middle thirds of the posterior border of SCM, where the **accessory nerve** and the lesser occipital, great auricular and transverse cervical nerves emerge.
- **Trapezius** forms the back of the posterior triangle; accessory nerve, elevates the shoulder.

> **CLINICAL PEARL** Accessory nerve injury in the posterior triangle causes a dropped shoulder, scapular malrotation and pain (shoulder syndrome) after neck dissection or node biopsy.

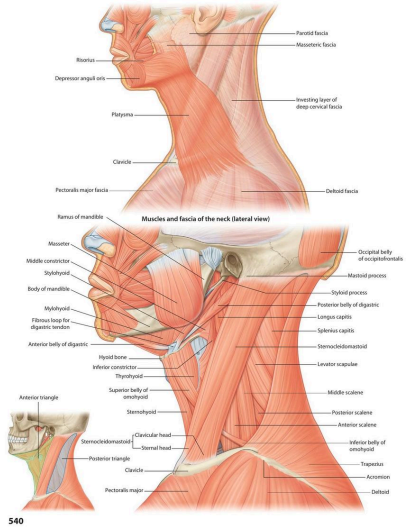
SOURCES DIFFER ERB'S POINT: ENT Steps puts XI and the cutaneous nerves at the junction of the upper and middle thirds; Doss and Gray's put the cutaneous nerves at the midpoint, with XI a little above. **XI INTO TRAPEZIUS:** Doss 5 cm above the clavicle; ENT Steps at the junction of its middle and lower thirds.



Superficial muscles of the neck
ENT Steps Anatomy, p. 264



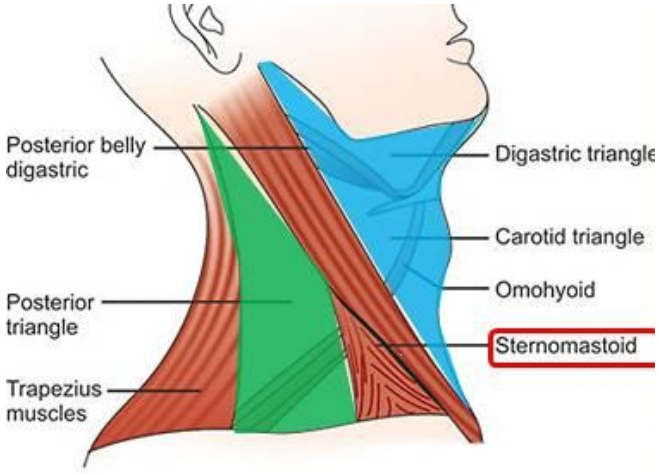
Muscles bounding the triangles of the neck
ENT Steps Anatomy, p. 266



Muscles and fascia of the neck (lateral view)
Gray's Atlas of Anatomy, 2nd ed., p. 540

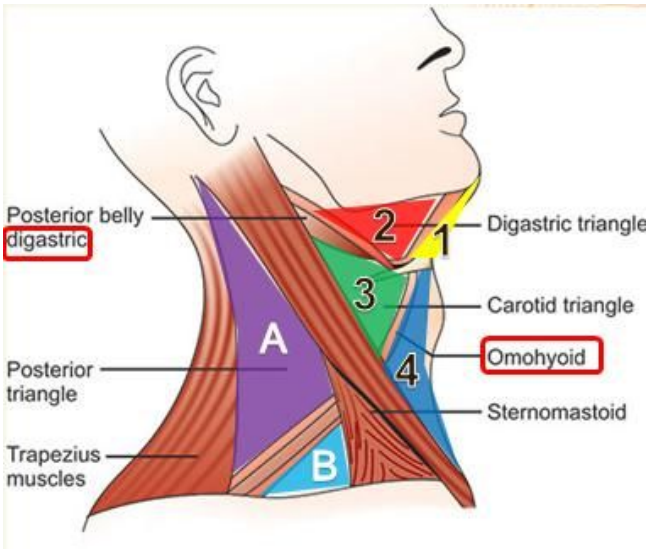
03 Triangles of the neck

- SCM divides each side into **anterior and posterior triangles**.
- **Anterior triangle** (SCM, lower border of mandible, midline) splits into: **submental** (anterior bellies of digastric and hyoid; nodes), **submandibular** (mandible and both bellies of digastric; submandibular gland, facial vessels, nodes, hypoglossal nerve), **carotid** (posterior belly of digastric, superior belly of omohyoid, SCM; carotid bifurcation, IJV, X, XII, ansa cervicalis, nodes), **muscular** (strap muscles, thyroid, larynx, trachea).
- **Posterior triangle** (SCM, trapezius, clavicle) splits by the inferior belly of omohyoid into **occipital** (accessory nerve, cervical plexus branches, nodes) and **supraclavicular** (subclavian vessels, brachial plexus trunks, transverse cervical vessels, nodes).



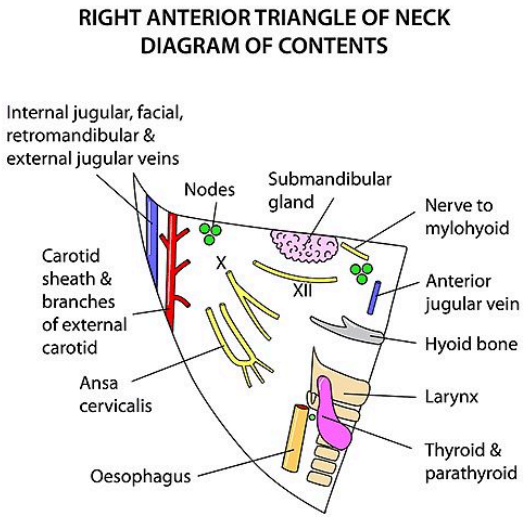
Triangles of the neck

ENT Steps Anatomy, p. 270



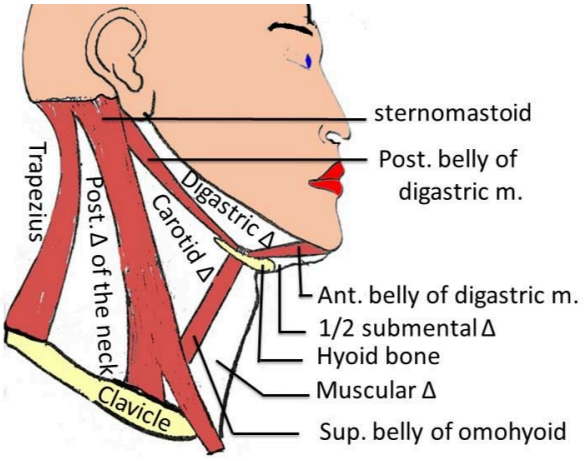
Subdivisions of the anterior and posterior triangles

ENT Steps Anatomy, p. 270



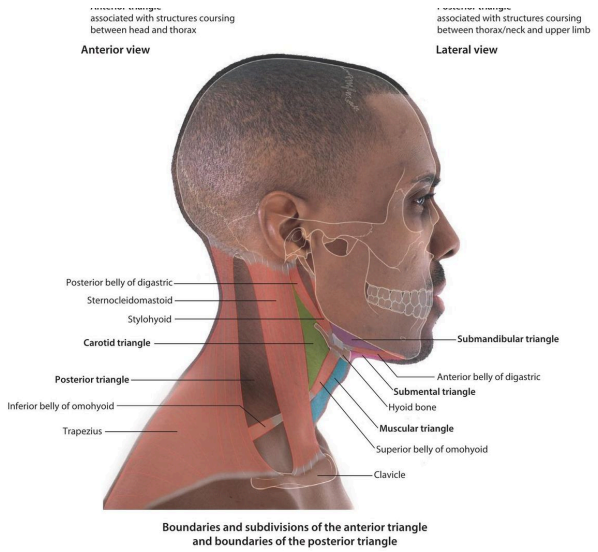
Contents of the anterior triangle

ENT Steps Anatomy, p. 271



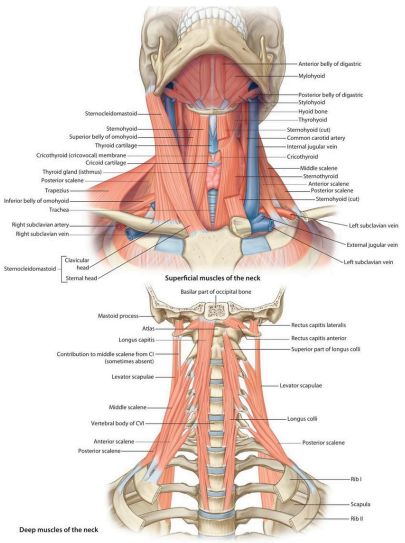
Triangles of the neck: boundaries

Dr. Sameh Doss handout, p. 112



Boundaries and subdivisions of the anterior and posterior triangles

Gray's Atlas of Anatomy, 2nd ed., p. 536

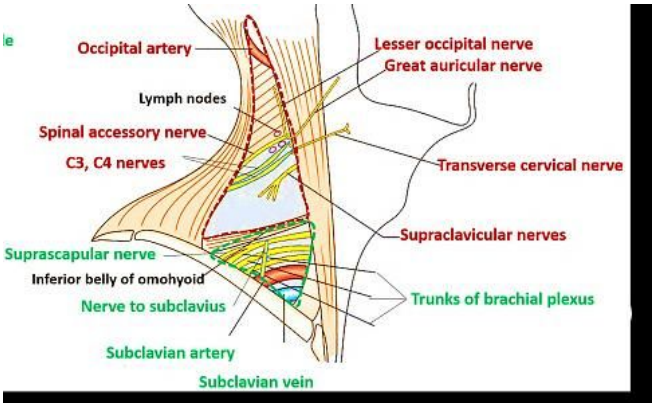


Superficial and deep muscles of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 541

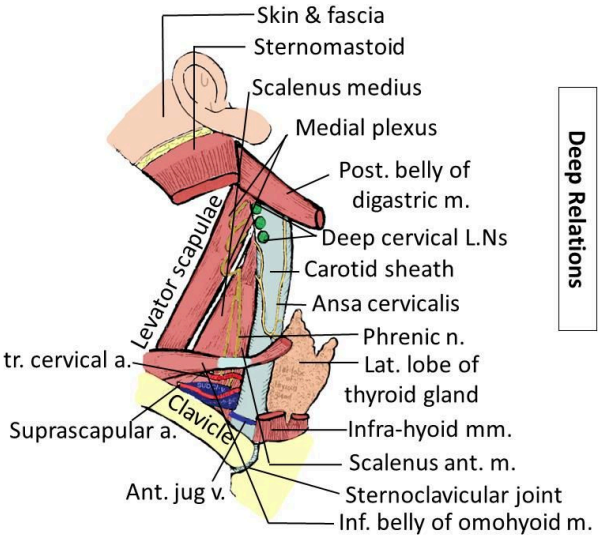
04 Posterior triangle in detail

- **Roof:** investing layer of deep cervical fascia. **Floor (muscles covered by prevertebral fascia):** splenius capitis, levator scapulae, scalenus medius and posterior, sometimes scalenus anterior.
- **Accessory nerve** crosses obliquely from Erb's point to the trapezius about 5 cm above the clavicle, lying in the roof.
- Lower part contains the **third part of the subclavian artery**, **trunks of the brachial plexus**, transverse cervical and suprascapular vessels, and the external jugular vein entering the subclavian vein.
- Phrenic nerve lies on scalenus anterior deep to the prevertebral fascia.



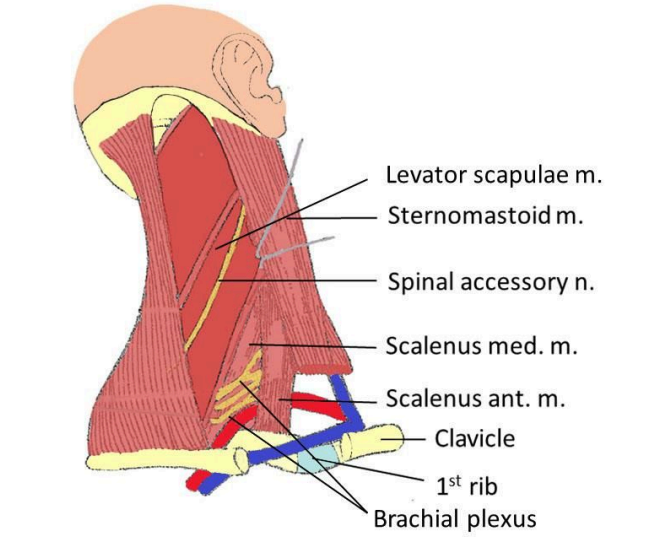
Contents of the posterior triangle

ENT Steps Anatomy, p. 272



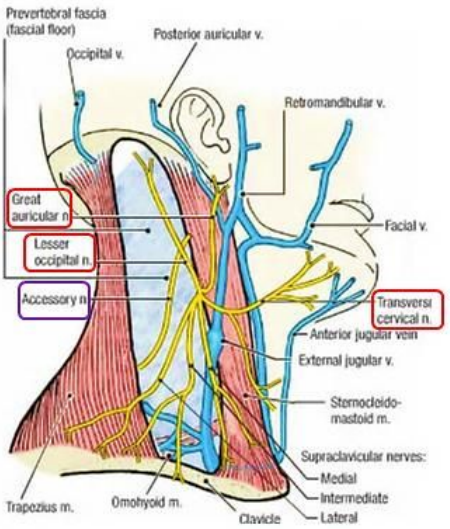
Deep relations of the posterior triangle

Dr. Sameh Doss handout, p. 113



Contents of the posterior triangle

Dr. Sameh Doss handout, p. 164



Nerves of the neck: cervical plexus and accessory nerve

ENT Steps Anatomy, p. 265

05 Deep cervical fascia

- **Superficial (investing) layer:** surrounds the neck and splits round **2 muscles** (SCM, trapezius) and **2 glands** (parotid, submandibular); forms the suprasternal space of Burns. Attached to the hyoid, mandible, mastoid, nuchal line, clavicle and sternum.

- **Middle (pretracheal, visceral) layer:** muscular part round the strap muscles; visceral part round the pharynx, larynx, trachea, oesophagus, thyroid and parathyroids; continues as buccopharyngeal fascia above and into the mediastinum below.
- **Deep (prevertebral) layer:** splits into **alar** fascia in front and **prevertebral** fascia behind. Prevertebral fascia covers the vertebral and scalene muscles, the phrenic nerve and brachial plexus, and extends as the axillary sheath.
- **Carotid sheath:** all three layers contribute; contains the **CCA/ICA, IJV and vagus**. The sympathetic chain lies behind it.
- The thyroid moves with swallowing because the pretracheal fascia attaches it to the larynx.

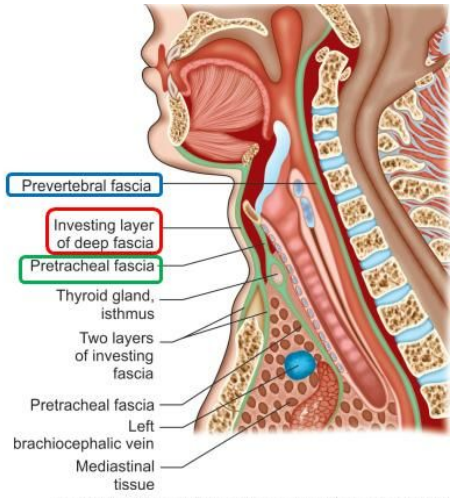


Fig. 3: Sagittal section of neck showing three layers of deep cervical fascia

Sagittal section: layers of deep cervical fascia

ENT Steps Anatomy, p. 274

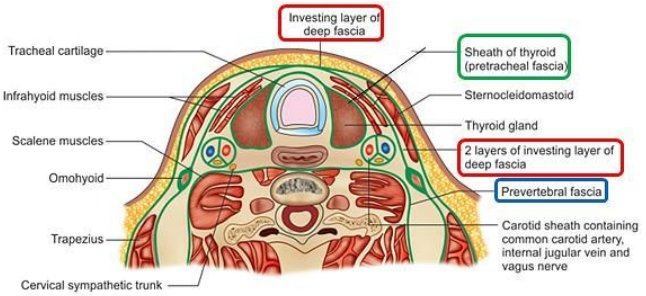
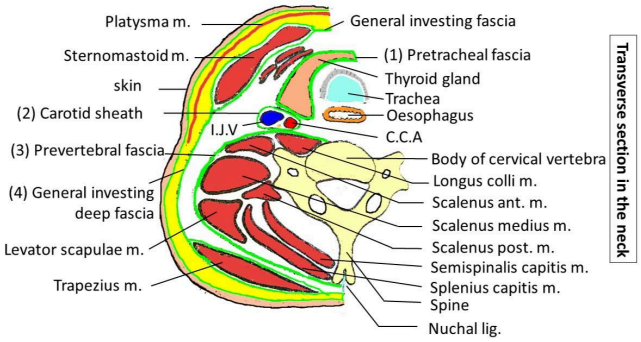


Fig. 4: Transverse section of lower part of neck showing three layers of deep cervical fascia

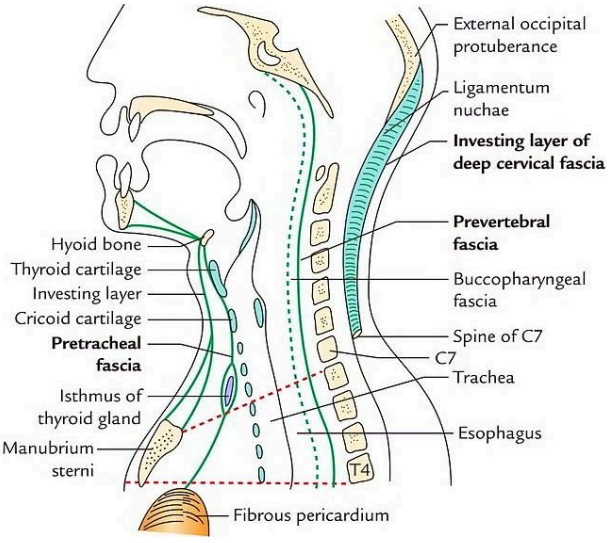
Transverse section: the three layers of deep cervical fascia

ENT Steps Anatomy, p. 274



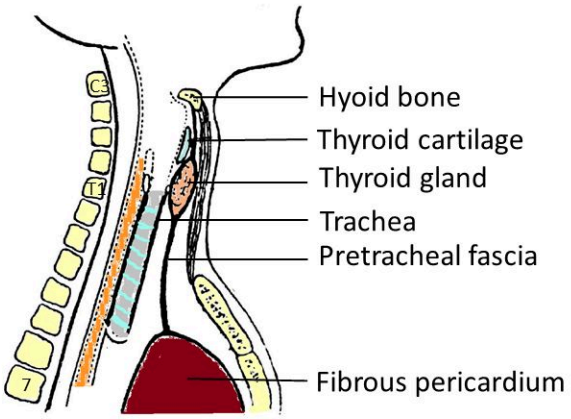
Deep cervical fascia in cross-section

Dr. Sameh Doss handout, p. 111



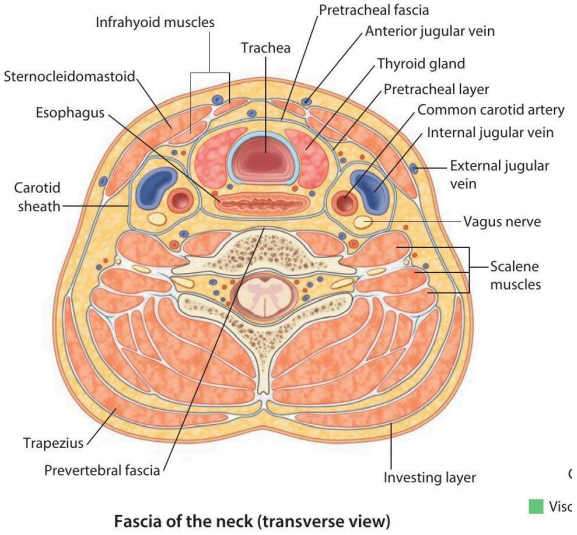
Deep cervical fascia in sagittal section

ENT Steps Anatomy, p. 275



Pretracheal fascia

Dr. Sameh Doss handout, p. 109



Fascia of the neck (transverse view)

Fascia of the neck (transverse view)

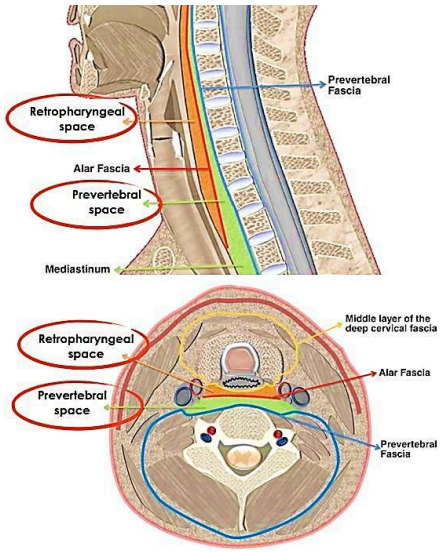
Gray's Atlas of Anatomy, 2nd ed., p. 538

06 Neck spaces

- **Retropharyngeal space:** between the visceral fascia (buccopharyngeal) and alar fascia, skull base to about **T2**.
- **Danger space:** between alar and prevertebral fascia, skull base to the **diaphragm**; infection spreads to the posterior mediastinum.
- **Prevertebral space:** behind the prevertebral fascia, the whole length of the spine.
- Also: submandibular (Ludwig's angina), parapharyngeal, peritonsillar and pretracheal spaces.

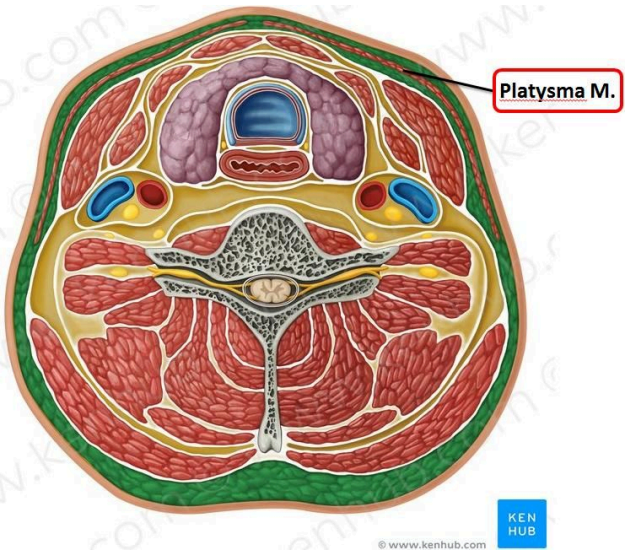
CLINICAL PEARL Infection in the danger space can reach the mediastinum quickly; widened prevertebral soft tissue on a lateral neck X-ray is the clue.

SOURCES DIFFER LOWER LIMIT OF THE RETROPHARYNGEAL SPACE: ENT Steps T2 (alar and visceral fascia fuse); Dhingra and Bansal about T4 (tracheal bifurcation); Scott-Brown T1–T2. **PREVERTEBRAL FASCIA:** ENT Steps to the coccyx; Doss to T3.



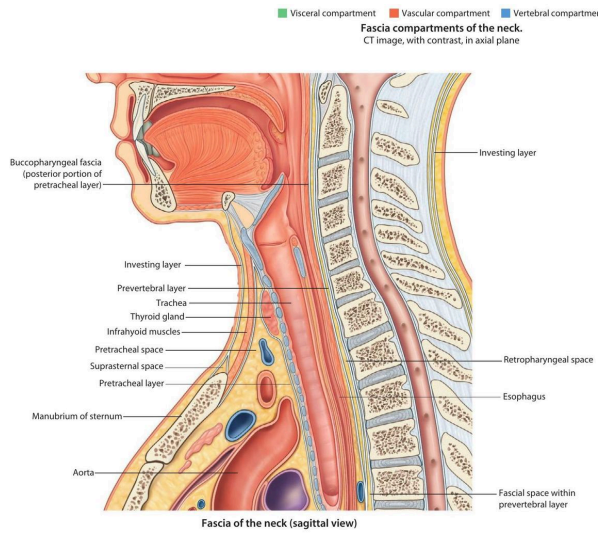
Retropharyngeal, danger and prevertebral spaces

ENT Steps Anatomy, p. 277



Cross-section of the neck at the thyroid level

ENT Steps Anatomy, p. 273



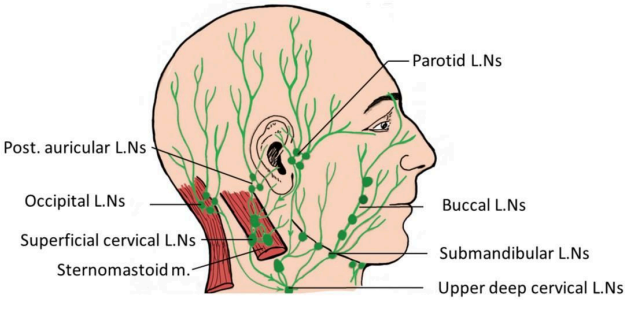
Fascia of the neck (sagittal view): pretracheal and retropharyngeal spaces

Gray's Atlas of Anatomy, 2nd ed., p. 538

07 Lymph node groups

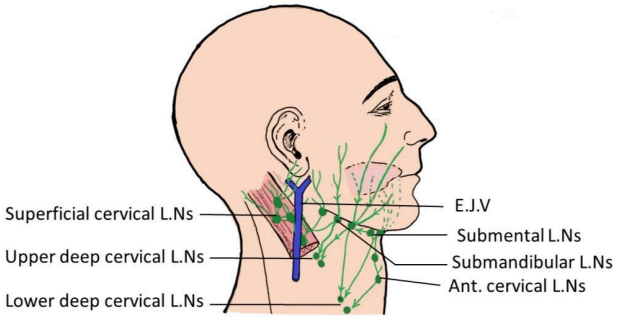
- **Superficial ring of the head:** occipital, mastoid, parotid (pre-auricular, intra-glandular), facial, submandibular, submental.
- **Deep cervical chains: internal jugular** (upper with the **jugulodigastric**, middle, lower with the **jugulo-omohyoid**), **spinal accessory**, **transverse cervical (supraclavicular)**.
- **Anterior (juxtavisceral):** prelaryngeal (Delphian), pretracheal, paratracheal (along the RLN).
- **Retropharyngeal:** lateral group with **Rouvière's node** at the atlas; not palpable.
- All lymph ends in the lower deep cervical nodes → jugular trunk → the jugulosubclavian junction (thoracic duct on the left).

> **CLINICAL PEARL** Left supraclavicular (Virchow's) node can be the first sign of abdominal or thoracic cancer via the thoracic duct.



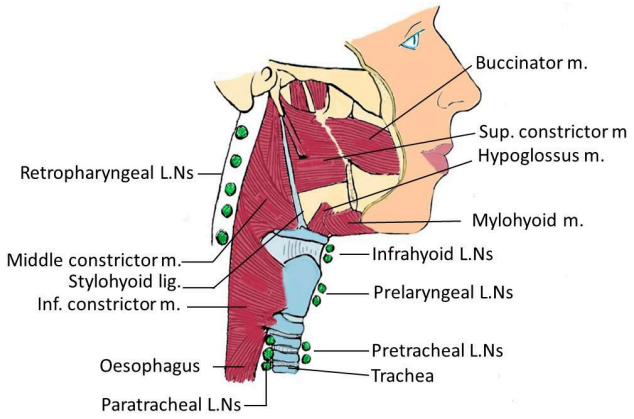
Superficial lymph nodes of the head

Dr. Sameh Doss handout, p. 195



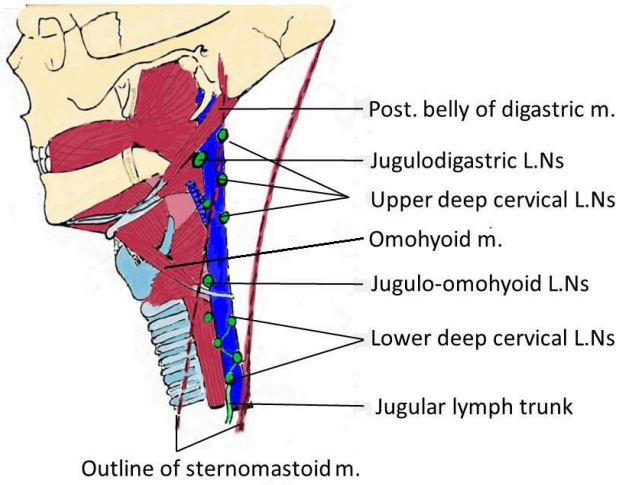
Superficial lymph nodes of the neck

Dr. Sameh Doss handout, p. 196



Deep circular lymph nodes of the head and neck

Dr. Sameh Doss handout, p. 197



Vertical group of deep cervical lymph nodes

Dr. Sameh Doss handout, p. 198

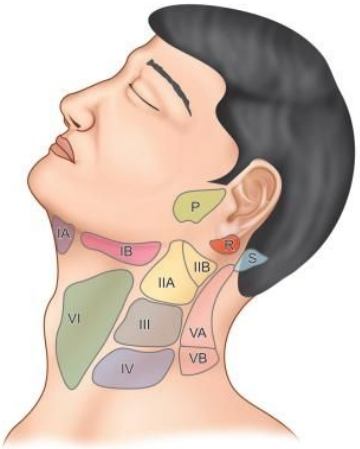
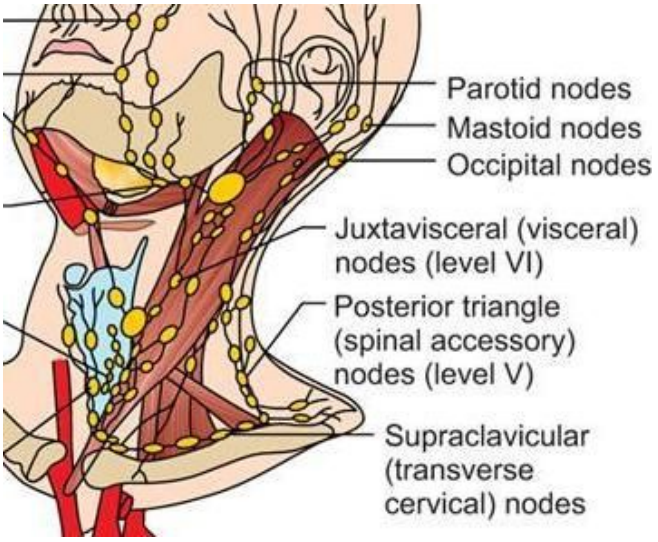


Fig. 5: Neck nodes regions. P, parotid preauricular; R, retroauricular, and S, suboccipital regions

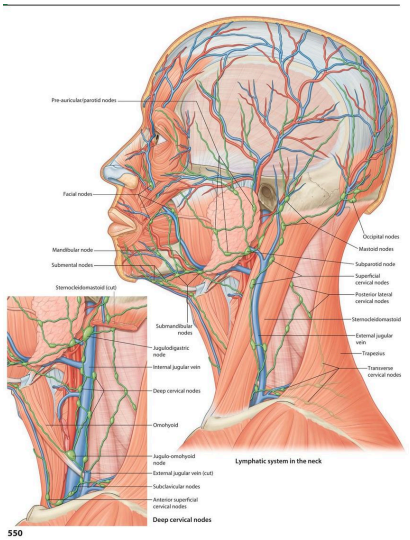
Lymph node regions of the head and neck

ENT Steps Anatomy, p. 284



Lymph node groups of the neck

ENT Steps Anatomy, p. 285



Lymphatic system in the neck: deep cervical nodes

Gray's Atlas of Anatomy, 2nd ed., p. 550

08 Lymph node levels

- **IA submental:** between the anterior bellies of digastric and above the hyoid. **IB submandibular.**
- **II upper jugular:** skull base to hyoid; **IIA** in front of (medial to) the accessory nerve, **IIB** behind (lateral to) it.
- **III middle jugular:** hyoid to the lower border of the cricoid (AAO-HNS). **IV lower jugular:** cricoid to the clavicle.
- **V posterior triangle: VA** above the cricoid level (accessory nodes), **VB** below (transverse cervical, supraclavicular).
- **VI central compartment:** hyoid to suprasternal notch, between the carotids (Delphian, pretracheal, paratracheal, perithyroid).
- **VII:** superior mediastinum below the suprasternal notch to the innominate artery.
- Not in levels: parotid, retroauricular, suboccipital, facial and retropharyngeal nodes.
- **Ho's triangle:** supraclavicular area between the medial and lateral ends of the clavicle and the neck–shoulder point; nodes here mean advanced nasopharyngeal cancer.

CLINICAL PEARL Neck dissections: radical = I–V plus SCM, IJV and accessory nerve. Modified radical keeps one or more of them. Selective: supraomohyoid I–III (oral cavity), lateral II–IV (larynx, pharynx), central VI (thyroid).

SOURCES DIFFER III/IV BOUNDARY: ENT Steps uses the upper border of the cricoid; AAO-HNS uses the lower border. **LEVEL VII:** the 2014 consensus (Grégoire) renames upper mediastinal nodes VIb and uses VII for retropharyngeal (VIIa) and retrostyloid (VIIb) nodes.

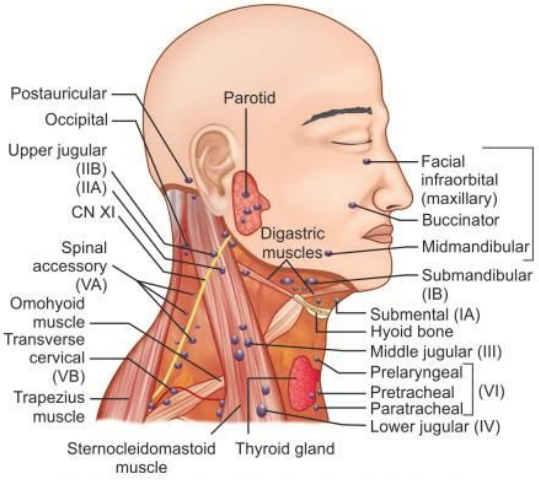
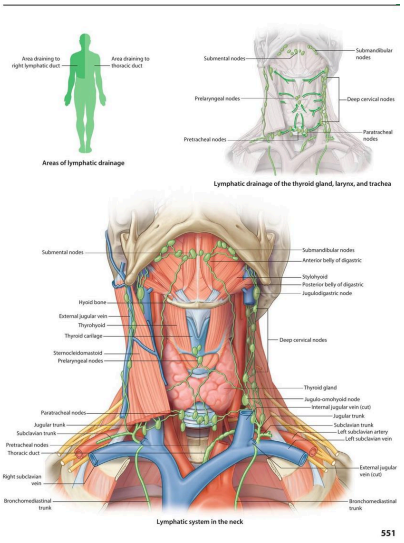


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

Lymph node levels of the neck
ENT Steps Anatomy, p. 285



Lymphatic drainage areas of the neck, thyroid, larynx and trachea
Gray's Atlas of Anatomy, 2nd ed., p. 551

09 Surgical anatomy: In theatre

- **Neck dissection landmarks:** marginal mandibular nerve (below the mandible), posterior belly of digastric (the "resident's friend": everything important lies deep to it), accessory nerve at the IJV and in the posterior triangle, hypoglossal nerve above the bifurcation, vagus in the sheath, phrenic and brachial plexus under the prevertebral fascia, thoracic duct on the left low neck.
- Stay above the **prevertebral fascia** to protect the phrenic nerve and brachial plexus.

- **Level IIB** dissection risks the accessory nerve; levels IIB and V are often omitted in node-negative necks.
- **Deep neck abscess drainage:** approach along the anterior border of SCM; the carotid sheath lies deep.

> **CLINICAL PEARL** Chyle leak: low-fat diet and pressure first; re-explore if over about 500 mL a day or persisting.

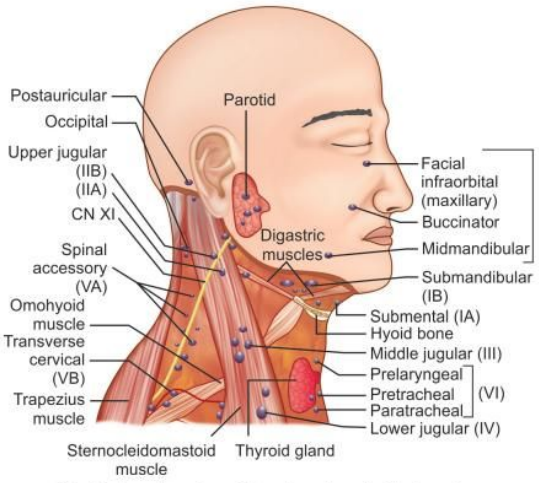
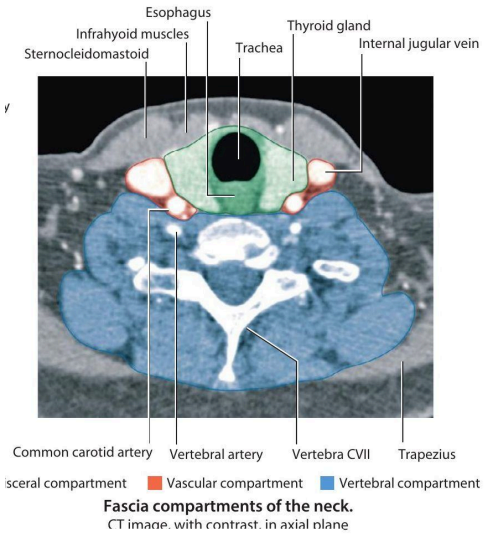


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

Lymph node levels of the neck
ENT Steps Anatomy, p. 285

10 Radiological anatomy: On imaging

- **Ultrasound** with FNA is first line for neck lumps; normal nodes are oval with a fatty hilum; round, hilumless, necrotic or calcified nodes are suspicious.
- **CT/MRI node criteria:** short axis over about 10 mm (11 mm at jugulodigastric), central necrosis, extranodal extension.
- **PET-CT** for unknown primary and recurrence; FDG uptake in Waldeyer's ring is normal.



Fascia compartments of the neck.
CT image, with contrast, in axial plane

Fascial compartments of the neck on contrast CT
Gray's Atlas of Anatomy, 2nd ed., p. 538

Tables

Triangles of the neck

Triangle	Boundaries	Main contents
Submental	Anterior bellies of digastric, hyoid	Submental nodes (IA)
Submandibular	Mandible, anterior and posterior bellies of digastric	Submandibular gland, facial vessels, IB nodes, XII, nerve to mylohyoid
Carotid	Posterior belly of digastric, superior belly of omohyoid, SCM	Carotid bifurcation, IJV, X, XII, ansa, deep cervical nodes
Muscular	Superior belly of omohyoid, SCM, midline	Strap muscles, thyroid, larynx, trachea
Occipital	SCM, trapezius, inferior belly of omohyoid	Accessory nerve, cervical plexus branches, nodes
Supraclavicular	SCM, inferior belly of omohyoid, clavicle	Subclavian artery (3rd part), brachial plexus trunks, transverse cervical vessels

Neck lymph node levels

Level	Boundaries	Main drainage
IA	Between anterior bellies of digastric, above hyoid	Chin, lower lip centre, anterior floor of mouth, tongue tip
IB	Submandibular triangle	Oral cavity, anterior nose, face, submandibular gland
IIA / IIB	Skull base to hyoid; in front of / behind XI	Oral cavity, nasopharynx, oropharynx, larynx, parotid
III	Hyoid to cricoid	Oral cavity, oropharynx, hypopharynx, larynx
IV	Cricoid to clavicle	Hypopharynx, larynx, thyroid, cervical oesophagus
VA / VB	Posterior triangle above / below cricoid	Nasopharynx, oropharynx, posterior scalp and neck skin
VI	Hyoid to suprasternal notch, between carotids	Thyroid, subglottis, cervical trachea and oesophagus
VII	Suprasternal notch to innominate artery	Thyroid, oesophagus, trachea

Types of neck dissection

Type	Levels removed	Structures sacrificed
Radical	I–V	SCM, IJV, accessory nerve
Modified radical type I	I–V	SCM and IJV (XI kept)
Modified radical type II	I–V	SCM (XI and IJV kept)
Modified radical type III (functional)	I–V	None
Supraomohyoid	I–III	None (oral cavity)
Lateral	II–IV	None (larynx, oro/hypopharynx)
Posterolateral	II–V + suboccipital, retroauricular	None (posterior scalp skin)
Central (anterior)	VI	None (thyroid, subglottis)
Extended radical	I–V plus extra nodes or structures	As radical plus others

Layers of deep cervical fascia

Layer	Encloses	Spaces formed
Investing (superficial)	SCM, trapezius, parotid, submandibular gland	Suprasternal space of Burns; roof of posterior triangle

Layer	Encloses	Spaces formed
Pretracheal – muscular	Strap muscles	—
Pretracheal – visceral	Pharynx, larynx, trachea, oesophagus, thyroid, parathyroids (buccopharyngeal fascia above)	Front wall of retropharyngeal space
Prevertebral – alar	—	Between retropharyngeal and danger space
Prevertebral proper	Vertebral and scalene muscles, phrenic nerve, brachial plexus	Floor of posterior triangle; prevertebral space
Carotid sheath (all three)	CCA/ICA, IJV, vagus	—

Branchial cleft anomalies

Type	Opening / site	Tract relations
1st (Work type I)	Near the EAC; ectoderm only	Duplication of the EAC, lateral to VII
1st (Work type II)	Angle of mandible; ectoderm + mesoderm	Through the parotid, close to the facial nerve
2nd (commonest)	Anterior border of SCM, lower third	Between ICA and ECA, over IX and XII, to the tonsillar fossa
3rd	Anterior border of SCM, lower	Behind the ICA, pierces thyrohyoid membrane above the internal laryngeal nerve, to the piriform sinus
4th	Usually left; piriform sinus apex	Below the superior laryngeal nerve; recurrent left neck abscess or thyroiditis

Vertebral levels in the neck

Level	Structure
C1	Atlas; Rouvière's node; nasopharynx level
C2	Soft palate, oropharynx begins
C3	Hyoid bone; epiglottis tip
C4	Upper border of thyroid cartilage; carotid bifurcation
C5	Vocal cords
C6	Cricoid; start of trachea and oesophagus; vertebral artery enters foramen; Chassaignac's tubercle
C7	Thoracic duct arch; vertebra prominens
T2	Lower end of retropharyngeal space
T4	Sternal angle; aortic arch
T5	Carina (T4–T5 in cadavers)
T6	Left main bronchus crosses the oesophagus (~25–27 cm from incisors)

Deep neck spaces

Space	Extent	Key contents / source
Retropharyngeal	Skull base to T2	Retropharyngeal nodes; children after URTI
Danger	Skull base to diaphragm	Loose fat; spread to mediastinum

Space	Extent	Key contents / source
Prevertebral	Skull base to coccyx	Vertebral (TB, osteomyelitis)
Parapharyngeal	Skull base to hyoid	Pre- and poststyloid compartments; dental, tonsil
Submandibular / sublingual	Floor of mouth above and below mylohyoid	Ludwig's angina (2nd and 3rd molars)
Peritonsillar	Tonsil capsule to superior constrictor	Quinsy

Suprahyoid and infrahyoid muscles

Muscle	Origin	Insertion	Nerve	Action
Stylohyoid	Base of styloid process	Body of hyoid (splits round digastric tendon)	VII	Pulls hyoid up and back
Digastric, anterior belly	Digastric fossa of mandible	Intermediate tendon to hyoid	Nerve to mylohyoid (V3)	Opens mouth; raises hyoid
Digastric, posterior belly	Mastoid notch	Intermediate tendon to hyoid	VII	Raises hyoid up and back
Mylohyoid	Mylohyoid line of mandible	Midline raphe and body of hyoid	Nerve to mylohyoid (V3)	Supports and raises floor of mouth
Geniohyoid	Inferior mental spines	Body of hyoid	C1 via XII	Pulls hyoid forward; depresses mandible
Sternohyoid	Manubrium and sternoclavicular joint	Body of hyoid	Ansa cervicalis (C1–C3)	Depresses hyoid
Omohyoid	Upper border of scapula near suprascapular notch	Lower border of hyoid body	Ansa cervicalis (C1–C3)	Depresses and fixes hyoid
Thyrohyoid	Oblique line of thyroid cartilage	Greater horn and body of hyoid	C1 via XII	Depresses hyoid or raises larynx
Sternothyroid	Back of manubrium	Oblique line of thyroid cartilage	Ansa cervicalis (C1–C3)	Pulls larynx down

Gray's Atlas. Infrahyoid (strap) muscles are retracted or divided in thyroidectomy; division high (near the larynx) spares their nerves, which enter low.

M Mnemonics

Investing fascia

Two muscles, two glands

SCM and trapezius; parotid and submandibular

Triangles of the anterior triangle

Some Silly Cats Meow

Submental, Submandibular, Carotid, Muscular

Erb's point nerves

LAST (with XI)

Lesser occipital, great Auricular, Supraclavicular, Transverse cervical, plus the accessory nerve

Levels

1 under the jaw, 2-3-4 down the jugular, 5 behind, 6 in front

Quick map of node levels

Q Rapid recall

1. **Contents of the carotid sheath?** Common/internal carotid artery, IJV and vagus nerve.

2. **What is Erb's point?** Junction of upper and middle thirds of the posterior border of SCM, where XI and the cutaneous cervical plexus branches emerge.
3. **Boundaries of the posterior triangle?** SCM in front, trapezius behind, middle third of the clavicle below.
4. **Which structures does the investing layer enclose?** Two muscles (SCM, trapezius) and two glands (parotid, submandibular).
5. **Boundaries of level II and how is it divided?** Skull base to hyoid; IIA in front of the accessory nerve, IIB behind it.
6. **Where is level VI?** Hyoid to suprasternal notch, between the carotid arteries.
7. **What is the danger space?** Between alar and prevertebral fascia, skull base to diaphragm.
8. **Levels removed in a supraomohyoid neck dissection?** I–III.
9. **What is Ho's triangle?** Supraclavicular area; nodes here mean advanced nasopharyngeal carcinoma.
10. **What is the vertebral level of the cricoid?** C6.
11. **Where is the IJV relative to SCM's heads?** In the gap between the sternal and clavicular heads, deep to the muscle.
12. **Nerve supply of SCM?** Accessory nerve (motor), C2–C3 proprioception (C2–C4 in ENT Steps).
13. **Floor of the posterior triangle?** Splenius capitis, levator scapulae, scalenus medius and posterior, covered by prevertebral fascia.
14. **Where does the accessory nerve enter trapezius?** About 5 cm above the clavicle.
15. **Which fascial layer forms the anterior wall of the retropharyngeal space?** The visceral (buccopharyngeal) layer of the pretracheal fascia.
16. **What is the suprasternal space of Burns?** A space formed by the splitting of the investing fascia above the sternum, containing the jugular arch.
17. **Boundaries of level III?** Hyoid to the cricoid, deep to SCM.
18. **Which levels are in a lateral neck dissection?** II–IV.
19. **Normal prevertebral soft tissue at C2 in adults?** Under about 7 mm.
20. **Which node is Virchow's node?** Left supraclavicular node near the thoracic duct.
21. **Which structures are sacrificed in a radical neck dissection?** SCM, IJV and accessory nerve, with levels I–V.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.